



Towards Classification of Typical and Atypical Disfluencies: A Self Supervised Representation Approach

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Abstract

This paper investigates the nuanced distinctions between typical and atypical speech disfluencies, focusing on features captured by self-supervised models. Typical disfluencies are natural, non-pathological irregularities in speech, while atypical disfluencies are linked to speech disorders like stuttering, characterized by more frequent and severe disruptions. Despite progress in automatic disfluency detection, limited research addresses the direct classification of these two types. This study leverages intermediate representations from four pre-trained models Wav2Vec2.0, HuBERT, WavLM, and TERA, to analyze and classify typical and stuttered disfluencies. The experiments utilize two novel Indian English datasets, IIITH-IED-E and IIITH-TISA, enabling a comprehensive analysis of disfluency patterns in a linguistically diverse context. Classification experiments with support vector machines (SVM) and convolutional neural networks (CNN) reveal that features from HuBERT's 5th layer—balancing low-level acoustic and high-level semantic information—achieve a peak F1 score of 0.97. These findings highlight the importance of intermediate layer representations of self-supervised models in distinguishing nuanced speech variations and contribute to robust and interpretable systems for speech disfluency classification.

Index Terms: Typical, Atypical, Wav2Vec2.0, WavLM, Hubert, TERA, SVM, CNN

1. Introduction

In the realm of human communication, the process of conveying and understanding messages is a remarkably complex endeavour, heavily influenced by a multitude of factors. From the subtle nuances of tone and intonation to the environmental context in which conversations take place, everyday communication demands considerable cognitive effort from both speakers and listeners. During spontaneous speech, individuals may experience difficulty retrieving words, necessitating sentence revisions or the insertion of pauses. Repetition of words or fragments is also a frequent occurrence. Studies suggest that approximately 6 out of every 100 fluent words are accompanied by some form of disfluency in typical speech[1][2]. Atypical speech represents a pronounced form of speech disfluency characterized by a significantly higher speech disruption rate than typical speech production. Stuttering is one such speech disorder characterised by atypical disfluencies, affecting about 70 million people worldwide, roughly 1% of the population[3]. In addition to the commonly recognized disfluencies, individuals who stutter may exhibit other types of speech disruptions that are distinguishable both quantitatively and qualitatively from those found in typical speech[4]. Understanding the disparities between these forms is crucial for early detection of stuttering in children, as they

may be mistakenly diagnosed with language development disfluency, leading to delayed intervention. Research indicates that children as young as two years old may begin to demonstrate an awareness of their stuttering condition, potentially leading to social consequences such as withdrawal from verbal communication. This can further complicate the diagnostic assessment for stuttering, which heavily relies on the individual's speech patterns[5][6]. This research paper builds on the work of [7], where handcrafted signal processing techniques were applied for feature extraction in disfluency classification, by employing self-supervised learning representations to enhance feature extraction and classification accuracy.

Interest in self-supervised representation learning has surged within paralinguistics, primarily due to the typically smaller datasets available compared to large-scale automatic speech recognition (ASR) datasets[8]. Self-supervised learning involves training a model in an unsupervised manner on extensive speech data, enabling it to capture meaningful audio representations directly from raw audio for specific downstream tasks. Notable pre-trained models, such as Wav2Vec2 [9], HuBERT [10], WavLM[11] and TERA [12] have demonstrated strong performance across a range of speech-related applications, including ASR, emotion recognition, speaker and language identification, and detecting vocal pathologies[8].

Self-supervised learning models differ based on aspects such as latent speech representation dimensions, input format, the nature of the self-supervision task, and the objective function used, leading each model to capture unique speech representation characteristics. To optimize for classification task, we experimented with different layers of multiple models to determine the most effective representation.

We observed that intermediate layers gave better accuracies than the final layer outputs. Intermediate layers retain critical spectral and temporal details without abstracting them to high-level semantics, which is essential for identifying subtle speech irregularities in conditions like disfluencies[13]. While early layers focus on raw, low-level features, task-oriented training fine-tunes the final layers to capture features most relevant to the output objective, often focusing on high-level semantic information rather than retaining finer-grained spectral or temporal details that may be essential for other tasks[14][15].

A major challenge in the pathological speech domain is the limited availability of datasets specifically designed for this classification task. Due to the practical and ethical challenges involved in collecting annotated speech data from children who stutter, we were unable to obtain a suitable pediatric dataset for this study. Therefore, we begin our investigation using adult speech data, which serves as a foundational step toward future extensions of this work to pediatric populations. Our experiments are conducted on two novel adult datasets introduced by

us in [7]

Our primary contributions are:

- **IIITH-TISA**: Collection, annotation, and curation of Indian English stammer dataset.
- **Extended IIITH-IED**: An expanded dataset of spontaneous non-pathological disfluencies with increased key events.
- **Comprehensive layer-wise analysis** of Wav2Vec2.0, HuBERT, WavLM, and TERA embeddings, both standalone and combined with acoustic features (openSMILE and eGeMAPS), for typical-atypical disfluency classification using SVM and CNN models.

2. Datasets

In collaboration with the Indian Stammering Association (TISA), an organization founded in 2008 to raise awareness and advocate for the rights of individuals who stammer across India, we gained access to their TISA Online Program Group (TOPG) 2.0 video series. This collection includes member-contributed recordings in Indian English on topics such as self-presentation, job interviews, and public speaking, designed for peer feedback, offering valuable resources for our research.

To develop a research dataset, we followed a rigorous pipeline: screening videos to ensure minimal background noise and removing any code-switching content or material involving active therapy. We then converted selected YouTube videos into audio-only WAV files, standardized to 16kHz, single-channel, and anonymized to protect privacy. This process produced 10 hours of speech data from 30 unique Persons Who Stutter (PWS). Using Audacity, we manually annotated each disfluency event with precise timestamps, following the SEP-28k[16] protocol to segment each clip into 3-second samples that contained the disfluency. Voice activity detection (VAD) was employed to trim to natural endpoints, and padding was added as needed. The final IIITH-TISA dataset includes 3,251 audio clips, each categorized by disfluency type, contributing valuable resources for stammering research in the Indian English context. In our analysis of the IIITH-IED dataset[17], the main source for typical disfluency patterns in this study, we identified a lower occurrence of certain disfluency types, particularly in Prolongation(elongated syllables) and all Repetition(repeated syllables/words/sentences) categories. To address this imbalance and improve dataset diversity, we expanded the annotations by including 30 additional speakers, resulting in an enhanced version we refer to as IIITH-IED-E.

TABLE 1 presents the statistics about the number of disfluency events in each of the datasets. Both datasets will be made accessible to other researchers upon request.

Table 1: *Disfluency Event Counts Across Three Datasets*

Disfluency Type	IIITH-TISA	IIITH-IED	IIITH-IED-E
Filled Pauses	626	1277	1277
Prolongations	313	71	293
Part-Word Repetitions	818	164	358
Phrase Repetitions	197	76	258
Word Repetitions	96	211	618
No Disfluencies	1201	2013	2013

3. Feature Extraction

This study utilizes self-supervised layer-wise feature representations from Wav2Vec2.0, WavLM, TERA, and HuBERT to capture subtle nuances in speech patterns for disfluency classi-

fication. A systematic comparison is conducted between traditional acoustic feature sets (openSMILE, eGeMAPS) and self-supervised representations. Additionally, the concatenation of acoustic features with the best-performing self-supervised representation, HuBERT-layer 5, is explored using SVM and CNN classifiers to assess its impact on classification performance.

We used the S3PRL speech toolkit along with its available pre-trained model weights to extract self-supervised speech representations[18].

Developed by Facebook AI Research, the Wav2Vec2.0 architecture comprises a multi-layer CNN encoder, a context network, and a quantization module. Pretraining employs a self-supervised approach akin to BERT’s masked language modeling for NLP[19], where consecutive time steps in the local encoder representations are randomly masked, and the model is trained to reconstruct the quantized representations at the contextualized encoder’s output. In this study, at each of the 24 transformer layers in the Wav2Vec2-LARGE model, 1024-dimensional feature vectors are extracted, which are then averaged across frames and used as input features for classification tasks.

WavLM is an advanced self-supervised learning model designed for effective speech representation, built upon the foundations of the wav2vec architecture but enhanced for various speech processing tasks. It employs a transformer-based encoder to process raw audio waveforms directly, extracting rich features for downstream tasks. The model architecture consists of a multi-layer transformer encoder that utilizes both masked prediction and next-token prediction as self-supervised tasks during training, enabling it to learn contextual information effectively from large-scale unlabeled audio datasets. WavLM supports various training objectives, including both masked language modeling and contrastive learning, which enhances its ability to capture diverse speech characteristics. The model also implements an attention mechanism that allows it to focus on relevant parts of the input audio, improving performance in tasks such as automatic speech recognition (ASR) and speaker recognition. Additionally, WavLM includes techniques for handling background noise and channel variability, making it robust for real-world applications. TERA, which stands for Transformer Encoder Representations from Alteration, is a novel self-supervised approach for speech pre-training. Unlike existing methods that typically rely on a singular auxiliary task, such as contrastive or autoregressive prediction, TERA employs alterations along three distinct axes: time, frequency, and magnitude. This technique allows the model to learn by reconstructing acoustic frames from their modified versions, utilizing a stochastic policy for diverse alterations.

The HuBERT model architecture is comprised of a multi-layer CNN encoder, a context network, a projection layer, and a code embedding layer. Pre-training involves k-means clustering (100 clusters) on MFCCs to generate targets, which are later refined over three iterations using features from transformer layers. The CNN encoder transforms speech into latent representations, which are projected to 1024 dimensions, masked, and passed through 24 transformer layers. Each 1024-dimensional features from HuBERT Large model are referred as HuBERT-N for layer-specific vectors.

While Wav2Vec2.0 and HuBERT, with their 24 transformer layers, produce 1024-dimensional feature vectors, WavLM (12 layers) and TERA (3-layer transformer encoder) yield 768-dimensional feature vectors.

4. Classification Framework

In this study, we employed two classifiers to distinguish between typical and atypical disfluencies: a Support Vector Machine (SVM) as the machine learning model and a Convolutional Neural Network (CNN) as the deep learning model. Parameter optimization for the SVM included experimentation with different kernels, ultimately selecting the Gaussian kernel, which achieved the highest classification accuracy. The CNN architecture was designed with three sequential convolutional layers, each followed by batch normalization and a leaky ReLU activation function. The output of these convolutional layers was then flattened and processed through two fully connected layers, where the final dense layer utilized a sigmoid activation function to perform binary classification between typical and atypical disfluencies. To evaluate classifier performance, we used accuracy and F1 score as metrics. While accuracy measures the overall rate of correct classifications, the F1 score, calculated as the harmonic mean of precision and recall, provides a balanced assessment especially useful for imbalanced data. This metric is critical for understanding the model’s performance in distinguishing between typical and atypical disfluencies, where false positives and false negatives have different implications. We used K-fold cross-validation with a 70/15/15 train/validation/test split, ensuring speaker exclusivity across sets to reliably assess model generalizability to unseen speakers.

4.1. Baseline Features for Comparison

The spectro-temporal feature set, Perceptually Enhanced ZTWCC, was evaluated against traditional features such as MFCC, PLPCC, and SFCC, as discussed in [7]. In the current study we compare Self Supervised Learning (SSL) feature representations with two widely-used acoustic feature sets—openSMILE, and eGeMAPS. The openSMILE set is a vector of 6,373 features, incorporating energy, spectral, and voicing-related descriptors. This comprehensive feature set is effective for examining nuanced speech abnormalities across various prosodic and spectral dimensions, providing an extensive basis for distinguishing atypical patterns in disfluent speech. In contrast, eGeMAPS, a more focused set of 88 features, includes specific acoustic metrics like pitch, jitter, formant frequencies, shimmer, and loudness, all of which have shown utility in prior studies for capturing physiological changes in vocal production. These features are particularly advantageous for assessing voice quality and prosodic shifts, which are crucial for characterising the subtleties of typical and atypical disfluency in speech. Additionally we concatenated the features of Hubert-layer5 that gave maximum F1 score amongst all the SSL features with eGeMAPS and openSMILE feature set using Canonical Correlation Analysis. CCA finds a linear transformation that projects both feature sets into a shared subspace where they are maximally correlated. In the following sections, features derived from self-supervised learning (SSL) models are referred to by the model name, with the suffix ‘-N’ denoting features extracted from the Nth layer of the corresponding model.

5. Results and discussions

Our experiments evaluated the effectiveness of SSL feature extraction techniques for classifying typical and atypical disfluencies. The classification focused on three distinct disfluency types: Repetition (including word, part-word, and phrase repetitions), Filled-Pause, and Prolongation.

Disfluent repetitions, common in English, often involve pauses and restarts, typically affecting unstressed function words [20, 21]. Stuttering can cause atypical repetitions due to disrupted laryngeal glottal abduction and prolonged vocal fold closure [22, 23]. Disfluent prolongations in stutterers lead to pitch accents on prolonged words and rushed subsequent speech, unlike in non-stutterers [6]. Filled pauses serve different functions in typical and atypical speech, with stutterers’ filled pauses aligning with disfluency onset, in typical speech, filled pauses mainly facilitate message planning, helping speakers maintain conversational control, capture the listener’s attention, or request assistance. The distinct characteristics of typical and atypical disfluencies in both production and manifestation are effectively captured by the initial layers of self-supervised models[24].

Table 2: *F1 Scores (in%) for SVM Classifier (PR.-Prolongation, Rep.-Repetition, F-P -Filled Pause) for typical vs atypical*

Features	Pr.	Rep.	F-P
Tera-2	90.39	93.14	91.95
Wav2Vec-3	95.01	95.82	95.84
WavLM-1	95.17	95.52	95.48
Hubert-5	97.67	97.95	97.89
eGeMAPS	91.37	92.85	91.88
openSMILE	90.67	90.85	91.89
HuBERT-5 + eGeMAPS	93.33	95.65	94.88
HuBERT-5 + openSMILE	92.11	92.95	92.20

The F1 score comparison between SVM and CNN classifiers (Tables 2 and 3) reveals how feature sets and disfluency categories influence performance, highlighting the interplay between feature representation and model architecture in achieving optimal classification outcomes.

5.1. SVM Classifier Observations

The SVM classifier achieves remarkable proficiency with HuBERT-5 features, with highest F1 scores across all disfluency types (Prolongation: 97.67%, Repetition: 97.95%, Filled-Pause: 97.89%). This consistency highlights HuBERT’s ability to leverage its deep, hierarchical structure to effectively capture intricate temporal and spectral patterns crucial for distinguishing subtle variations in disfluency. Wav2Vec-3 and WavLM-1, while not outperforming HuBERT-5, also display good accuracy, with F1 scores consistently greater than 95%, highlighting their potential as robust alternatives in self-supervised learning contexts.

Traditional acoustic features like eGeMAPS and openSMILE achieve 90–92% accuracy but fall short of HuBERT-5, which better captures fine-grained disfluency details. Combining HuBERT-5 with eGeMAPS improves performance over eGeMAPS alone but remains inferior to HuBERT-5, highlighting its superior ability to capture disfluency-specific nuances.

5.2. CNN Classifier Observations

The CNN classifier exhibits considerable variability, with HuBERT-4 features yielding the highest F1 scores across disfluency types (Prolongation: 94.51%, Repetition: 95.71%, Filled-Pause: 95.84%). While CNNs can process high-dimensional inputs, their performance suggests a limited capacity to exploit the hierarchical and prosodic structure embedded in HuBERT representations. Performance further declines with hand-crafted features like eGeMAPS and openSMILE, likely due

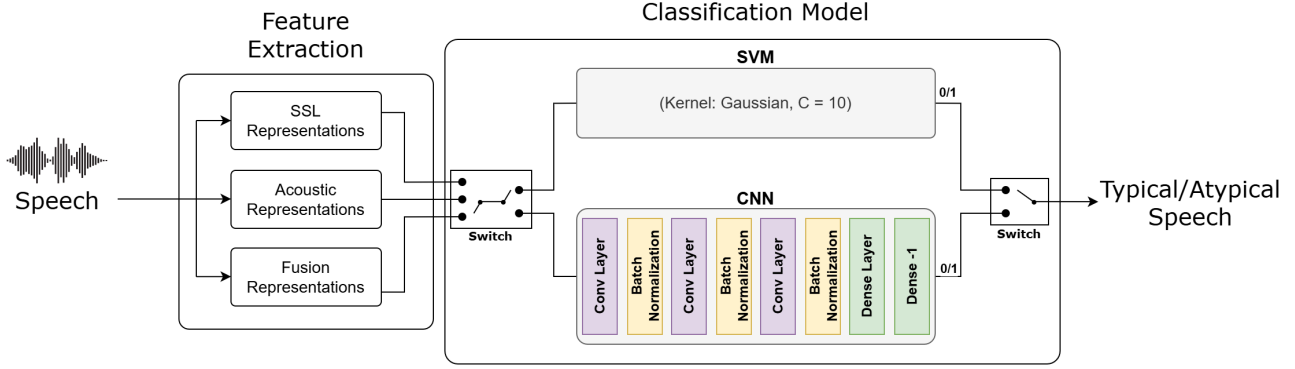


Figure 1: A schematic block diagram for classification of Typical and Atypical disfluencies

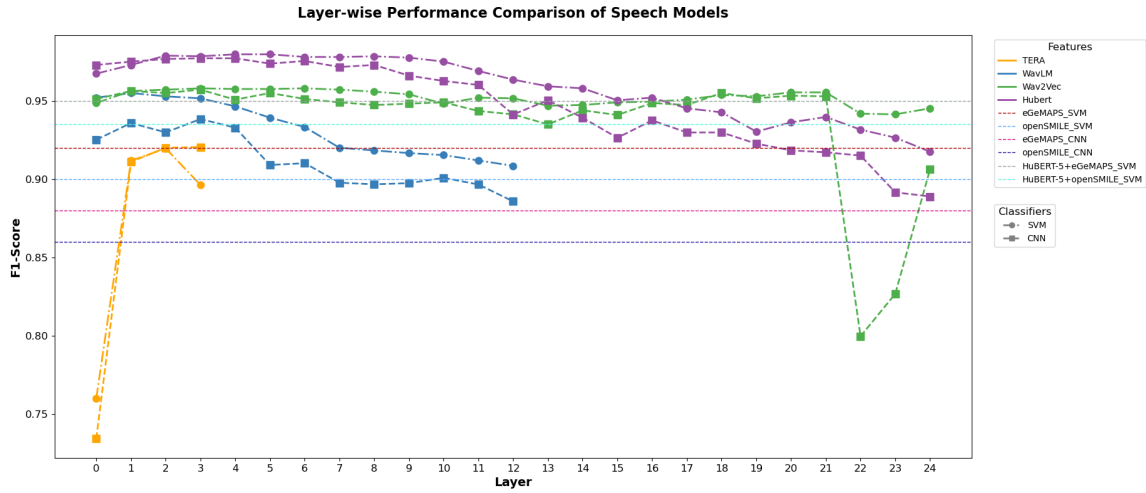


Figure 2: $F1$ scores of layer-wise SSL model features across layer depths. The graph illustrates the variation in classification performance, highlighting layers that capture disfluency characteristics most effectively.

to the CNN’s reliance on spatial locality, which is insufficient for modeling the temporal dynamics and suprasegmental cues critical to disfluency classification. In contrast, SVMs achieve strong results with HuBERT-5 features, benefiting from margin-based decision boundaries that align well with the abstract, multiscale representations produced by self-supervised learning.

HuBERT outperforms models like Wav2Vec2.0, TERA, and WavLM in disfluency classification due to its stronger prosody modeling. Its use of unsupervised clustering and masked prediction enables learning of suprasegmental features like pitch, rhythm, and pauses, crucial for distinguishing typical from atypical disfluencies. Unlike models focused on local acoustics, HuBERT captures longer-range prosodic patterns essential for this task. As shown in Figure 2, intermediate to deeper HuBERT layers, particularly layer 5, consistently deliver optimal classification performance.

6. Conclusion

This research introduces two key datasets: IITH-TISA, India’s first Indian English stammer corpus, and the expanded IITH-IED-E for typical disfluencies. By leveraging self-supervised learning (SSL) models alongside traditional prosodic features in SVM and CNN frameworks, this study explores the classifica-

Table 3: $F1$ Scores (in%) for CNN Classifier for typical vs atypical.

Features	Pr.	Rep.	F-P
Tera-3	88.21	93.21	92.03
Wav2Vec-2	95.01	95.65	95.73
WavLM-3	87.75	93.94	93.84
Hubert-4	94.51	95.71	95.84
eGeMAPS	87.37	87.85	85.88
openSMILE	88.67	85.85	87.89
HuBERT-5 + eGeMAPS	89.33	90.15	88.88
HuBERT-5 + openSMILE	88.11	85.95	89.20

tion of typical and atypical disfluencies. Results show that SSL-derived features, particularly from HuBERT-5, capture nuanced disfluency characteristics, surpassing traditional features like eGeMAPS and openSMILE. Future explorations may probe the potential for model ensembles or advanced multi-modal feature fusion techniques to elevate CNN performance on prosodic data, thus broadening the practical applicability of these models.

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